



Maths By Gagan Pratap

Trigonometry Sheet-2

Maths Special Batch
By Gagan Pratap

1. Which of the following is NOT a correct formula?

निम्नलिखित में से कौन सा सही सूत्र नहीं है?

(SSC SELECTION POST XII HIGHER SECONDARY LEVEL)

- A) $\frac{1+\tan^2 A}{\tan^2 A} = \sin^2 A$
 B) $\sec A \sin A = \tan A$
 C) $\operatorname{cosec}^2 A - \cot^2 A = 1$
 D) $\cos^2 A \cot^2 A = \cot^2 A - \cos^2 A$

2. If $\cot A = k$, then $\sin A$ is equal to: (Presume that A is an acute angle)

यदि $\cot A = k$ है, तो $\sin A$ का मान बताइए (माल लें कि A न्यून कोण है)

- (a) $\frac{1}{\sqrt{1+k^2}}$ (b) $\frac{k^2}{\sqrt{1+k^2}}$ (c) $\frac{1}{k}$ (d) $-\frac{1}{k}$

3. If $21\tan\theta = 20$, then $(1 + \sin\theta + \cos\theta) : (1 - \sin\theta + \cos\theta) = ?$

यदि $21\tan\theta = 20$ है, तो $(1 + \sin\theta + \cos\theta) : (1 - \sin\theta + \cos\theta) = ?$

- (a) 5 : 2 (b) 3 : 1 (c) 2 : 1 (d) 7 : 3

4. If $x = 8(\sin\theta + \cos\theta)$ & $y = 9(\sin\theta - \cos\theta)$, then find the value of $\frac{x^2}{8^2} + \frac{y^2}{9^2}$?

यदि $x = 8(\sin\theta + \cos\theta)$ & $y = 9(\sin\theta - \cos\theta)$ है, तो $\frac{x^2}{8^2} + \frac{y^2}{9^2}$ का मान ज्ञात करें?

- A) 4
 B) 6
 C) 8
 D) 2

5. If $x = a\sin\theta - b\cos\theta$, $y = a\cos\theta + b\sin\theta$, then which of the following is true?

- a) $\frac{x^2}{y^2} + \frac{a^2}{b^2} = 1$ b) $x^2 + y^2 = a^2 - b^2$ c) $x^2 + y^2 = a^2 + b^2$ d) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

6. If $x\cos A - y\sin A = 1$ and $x\sin A + y\cos A = 4$, then the value of $17x^2 + 17y^2$ is:

यदि $x\cos A - y\sin A = 1$ and $x\sin A + y\cos A = 4$ है तो $17x^2 + 17y^2$ का मान क्या है?

- a) 7 b) 49 c) 0 d) 289

7. If $x = 4\cos A + 5\sin A$ and $y = 4\sin A - 5\cos A$, then the value of

$x^2 + y^2$ is: -

यदि $x = 4\cos A + 5\sin A$ और $y = 4\sin A - 5\cos A$ हो, तो $x^2 + y^2$ का मान ज्ञात कीजिए

- a) 25 b) 16 c) 0 d) 41

8. If $6x = \sin\theta$ and $\frac{6}{x} = \cos\theta$, then the value of $8\left(x^2 + \frac{1}{x^2}\right)$ is:

यदि $6x = \sin\theta$ और $\frac{6}{x} = \cos\theta$ है तो $8\left(x^2 + \frac{1}{x^2}\right)$ का मान ज्ञात करें।

- (a) 2 (b) $\frac{2}{9}$ (c) 4 (d) 1

9. $x = a + a\sin\alpha \cdot \cos\beta$, $y = b(1 + \sin\alpha \cdot \sin\beta)$, $z = c \cdot \cos\alpha$, then $\left(\frac{x-a}{a}\right)^2 + \left(\frac{y-b}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = ?$

- a) 3 b) 1 c) -1 d) 0

10. $(1 - \sin A \cdot \cos A)(\sin A + \cos A) = ?$

- a) $\sin^2 A - \cos^2 A$ b) $\sin^3 A + \cos^3 A$ c) $\cos^2 A - \sin^2 A$ d) 0

11. If $\sin(A+B) = 1$ and $\cos(A-B) = \frac{\sqrt{3}}{2}$, $A+B \leq 90^\circ$ and $A > B$, then the value of $\frac{5\sin^2 B + 4\tan^2 A}{2\sin B \cos A}$ is:

यदि $\sin(A+B) = 1$ और $\cos(A-B) = \frac{\sqrt{3}}{2}$ है, $A+B \leq 90^\circ$ और $A > B$ है, तो $\frac{5\sin^2 B + 4\tan^2 A}{2\sin B \cos A}$ का मान ज्ञात करें।

- (a) 20

- (b) $26\frac{1}{2}$

- (c) 18



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(d) $16\frac{1}{2}$

12. If $\sec x + \cos x = 5/2$, where x lies between 0° and 90° , then what is the value of $\sin^2 x$?

यदि $\sec x + \cos x = 5/2$, जहाँ x , 0° और 90° के मध्य स्थित है, तो $\sin^2 x$ का मान क्या है?

(a) $3/4$

(b) $1/2$

(c) 1

(d) $1/4$

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13. If $t = \cos 79^\circ$, then what is $\operatorname{cosec} 79^\circ (1 - \cos 79^\circ)$ equal to?

यदि $t = \cos 79^\circ$, तो $\operatorname{cosec} 79^\circ (1 - \cos 79^\circ)$ किसके बराबर है? (CDS-1 2024)

A) $\sqrt{\frac{1+t}{1-t}}$

B) $\frac{t}{\sqrt{1-t^2}}$

C) $\frac{\sqrt{1-t^2}}{t}$

D) $\sqrt{\frac{1-t}{1+t}}$

14. If $16 \sec^2 \theta - 40 \sec \theta + 25 = 0$ and θ is an acute angle, then what will be the value of $\tan \theta$?

यदि $16 \sec^2 \theta - 40 \sec \theta + 25 = 0$ और θ एक न्यून कोण है, तो $\tan \theta$ का मान कितना होगा?

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[A] $\frac{4}{3}$

[B] $\frac{3}{4}$

[C] $\frac{4}{5}$

[D] $\frac{3}{5}$

15. If $8 \sin^2 \theta + 2 \cos \theta = 5$, $0^\circ < \theta < 90^\circ$, then the value of $\tan^2 \theta + \sec^2 \theta - \sin^2 \theta$ will be:

यदि $8 \sin^2 \theta + 2 \cos \theta = 5$ है, $0^\circ < \theta < 90^\circ$ है, तो $\tan^2 \theta + \sec^2 \theta - \sin^2 \theta$ का मान ज्ञात करें।

(a) $\frac{305}{144}$

(b) $\frac{431}{144}$

(c) $\frac{23}{29}$

(d) $\frac{153}{72}$

16. If $3 \cos \alpha + 10 \sin^2 \alpha = 6$ ($0 < \alpha < 90^\circ$), Then find $\cot \alpha = ?$

(a) $\frac{4}{5}$ (b) $\frac{4}{3}$ (c) $\frac{1}{4}$ (d) $\frac{3}{2}$

17. If $6 - 4 \sin^2 \theta = 7 \sin \theta \cdot \cos \theta$ then find $\cot \theta = ?$

(a) $\frac{2}{3}$ (b) $\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{3}{2}$

18. $(1 + \sin \alpha)(1 + \sin \beta)(1 + \sin \gamma) = (1 - \sin \alpha)(1 - \sin \beta)(1 - \sin \gamma)$

Then value of each side is:

(a) $\pm \sin \alpha \sin \beta \sin \gamma$ (b) $\pm \sec \alpha \cdot \sec \beta \sec \gamma$

(c) $\pm \cos \alpha \cdot \cos \beta \cdot \cos \gamma$ (d) $\pm \tan \alpha \cdot \tan \beta \cdot \tan \gamma$



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19. $(1 + \cos \alpha)(1 - \sin \beta)(1 + \cos \gamma)(1 - \sin \delta) = (1 - \cos \alpha)(1 + \sin \beta)(1 - \cos \gamma)(1 + \sin \delta)$
then value of each term is =?

(a) $\pm \sin \alpha \sin \beta \sin \gamma \sin \delta$ (b) $\pm \sec \alpha \cdot \sec \beta \sec \gamma \sec \delta$
(c) $\pm \cos \alpha \cdot \sin \beta \cdot \cos \gamma \sin \delta$ (d) $\pm \sin \alpha \cdot \cos \beta \cdot \sin \gamma \cos \delta$

20. If $\sin \alpha + \cos \beta = 2$, then $\cos \alpha + \sin \beta = ?$

(a) 0 (b) 1 (c) 2 (d) -1

21. If $\cos A + \cos B + \cos C = 3$, then what is the value of $\sin A + \sin B + \sin C$?

यदि $\cos A + \cos B + \cos C = 3$, तो $\sin A + \sin B + \sin C$ का मान क्या है?

(a) 1 (b) 2 (c) 0 (d) -1

22. $\left(\frac{1-\tan \theta}{1-\cot \theta}\right)^2 + 1 = ?$

$\left(\frac{1-\tan \theta}{1-\cot \theta}\right)^2 + 1$ बराबर है :

a) $\sec^2 \theta$ b) $\cos^2 \theta$
c) $\sin^2 \theta$ d) $\operatorname{cosec}^2 \theta$

23. If $\sin \theta + \operatorname{cosec} \theta = 2$, Then $\sin^{100} \theta + \operatorname{cosec}^{100} \theta = ?$

(a) 0 (b) 1 (c) 2 (d) 4

24. If $A = 2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta)$ then the value of 3α such that $\cos \alpha = \sqrt{\frac{3+A}{5+A}}$ is:

यदि $A = 2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta)$ है तो $\cos \alpha = \sqrt{\frac{3+A}{5+A}}$ करने के लिए 3α का मान ज्ञात करें

(a) 45° (b) 135° (c) 180° (d) 90°

25. If $1 + \cos x + \cos^2 x + \cos^3 x + \dots \infty$ terms = $4 + 2\sqrt{3}$, then $x = ?$

(a) 15° (b) 30° (c) 45° (d) 60°

26. Find the value of $\sqrt{\frac{1-\sin 3\theta}{1+\sin 3\theta}}$?

$\sqrt{\frac{1-\sin 3\theta}{1+\sin 3\theta}}$ का मान ज्ञात कीजिये?

- A) $\sec 3\theta - \tan 3\theta$
B) $(\sec 3\theta - \tan 3\theta)^2$
C) $(\sec 3\theta + \tan 3\theta)^2$
D) $\sec 3\theta + \tan 3\theta$

27. $\frac{\cos \theta}{\sec \theta - 1} + \frac{\cos \theta}{\sec \theta + 1}$ is equal to:

$\frac{\cos \theta}{\sec \theta - 1} + \frac{\cos \theta}{\sec \theta + 1}$ का मान _____ के बराबर है।

(a) $2\sin^2 \theta$ (b) $2\sec^2 \theta$ (c) $2\cot^2 \theta$ (d) $2\cos^2 \theta$

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28. The value of $\sin 2\theta (\tan \theta + \cot \theta)$ is:

$\sin 2\theta (\tan \theta + \cot \theta)$ का मान क्या होगा?

(a) $\frac{2}{3}$
(b) 1
(c) $\frac{1}{2}$
(d) 2



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29. The value of $\frac{\tan^2 \theta - \sin^2 \theta}{2 + \tan^2 \theta + \cot^2 \theta}$ is : —

$\frac{\tan^2 \theta - \sin^2 \theta}{2 + \tan^2 \theta + \cot^2 \theta}$ का मान क्या है?

- a) $\operatorname{cosec}^6 \theta$ b) $\cos^4 \theta$ c) $\sin^6 \theta$ d) $\sec^4 \theta$

30. The value of $\sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} \div \frac{\sin \theta}{1 + \cos \theta}$ is equal to:

$\sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} \div \frac{\sin \theta}{1 + \cos \theta}$ का मान बराबर है :

- a) 1 b) $\operatorname{cosec} \theta$
c) $\sec \theta$ d) $\frac{1}{2}$

31. The value of $\frac{2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta)}{\cos^4 \theta - \sin^4 \theta - 2\cos^2 \theta}$ is:

$\frac{2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta)}{\cos^4 \theta - \sin^4 \theta - 2\cos^2 \theta}$ का मान है :

- a) -2 b) 1
c) -1 d) 2

32. The value of $\frac{\cos^6 \theta + \sin^6 \theta + 3\sin^2 \theta \cos^2 \theta}{\operatorname{cosec} \theta \sec \theta (\sin \theta + \cos \theta - 1)(\sin \theta + \cos \theta + 1)}$ is:

$\frac{\cos^6 \theta + \sin^6 \theta + 3\sin^2 \theta \cos^2 \theta}{\operatorname{cosec} \theta \sec \theta (\sin \theta + \cos \theta - 1)(\sin \theta + \cos \theta + 1)}$ का मान ज्ञात कीजिए।

- (a) 1 (b) 2
(c) $\frac{1}{2}$ (d) 3

33. The expression $\frac{(1 - 2\sin^2 \theta \cos^2 \theta)(\cot \theta + 1)\cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta)\operatorname{cosec} \theta}$, $0^\circ < \theta < 90^\circ$, is equal to:

व्यंजक $\frac{(1 - 2\sin^2 \theta \cos^2 \theta)(\cot \theta + 1)\cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta)\operatorname{cosec} \theta}$ का मान बताइए जहाँ $0^\circ < \theta < 90^\circ$ है।

- (a) $\cos^2 \theta$ (b) $-\sin^2 \theta$
(c) $\sec^2 \theta$ (d) $-\sec^2 \theta$

34. $\frac{\sin^2 \theta}{\cos \theta (1 + \cos \theta)} + \frac{1 + \cos \theta}{\cos \theta} = ?$

- (a) $2 \cos \theta$ (b) $\sec \theta$
(c) $2 \sec \theta$ (d) $\operatorname{cosec} \theta$

35. $\frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta - 1} + \frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta + 1} - \tan^2 \theta$, $0^\circ < \theta < 90^\circ$, is equal to:

$\frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta - 1} + \frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta + 1} - \tan^2 \theta$, $0^\circ < \theta < 90^\circ$, का मान ज्ञात करें।

- (a) $2\sec^2 \theta$
(b) $\sec^2 \theta + 1$
(c) $\sec^2 \theta$
(d) $1 - \tan^2 \theta$

36. What is the value of $\sin A \left(1 + \frac{\sin A}{\cos A}\right) + \cos A \left(1 + \frac{\cos A}{\sin A}\right)$?

$\sin A \left(1 + \frac{\sin A}{\cos A}\right) + \cos A \left(1 + \frac{\cos A}{\sin A}\right)$ का मान क्या है?

- A) $\sec A + \operatorname{cosec} A$
B) $\sin A + \cos A$
C) $\sin A - \cos A$
D) $\sec A - \operatorname{cosec} A$

37. $\left(\frac{\operatorname{cosec} A}{\cot A + \tan A}\right)^2 = ?$

- a) $2 \cos^2 A$ b) $1 - \sin^2 A$
c) $\sec^2 A$ d) $\sec A \cdot \tan A$



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38. value of $\cos^2(90^\circ - \theta) - \left[\frac{\{\cos(90^\circ - \theta)\cos\theta\}}{\cot\theta} \right] ?$

$\cos^2(90^\circ - \theta) - \left[\frac{\{\cos(90^\circ - \theta)\cos\theta\}}{\cot\theta} \right]$ का सरलीकृत मान क्या है?

- (a) 4
(b) 2
(c) 0
(d) 1

39. Solve $\frac{(1 - \sin\theta + \cos\theta)^2}{2}$:

- (a) $(1 - \cos\theta)(1 - \sin\theta)$ (b) $(1 + \cos\theta)(1 - \sin\theta)$
(c) $(1 + \cos\theta)(1 + \sin\theta)$ (d) $(1 - \cos\theta)(1 + \sin\theta)$

40. If $\cos A$, $\sin A$, $\cot A$ are in geometric progression, then the value of $\tan^6 A - \tan^2 A$ is:

अगर $\cos A$, $\sin A$, $\cot A$ गुणोत्तर श्रेणी में हैं तो $\tan^6 A - \tan^2 A$ का मान क्या होगा।

- (a) $\frac{1}{2}$ (b) 3 (c) $\frac{1}{3}$ (d) 1

41. If $\sin x + \cos x = c$, then $\sin^6 x + \cos^6 x$ is equal to:

- a) $\frac{1+6c^2-3c^4}{16}$ b) $\frac{1+6c^2-3c^4}{4}$ c) $\frac{1+6c^2+3c^4}{16}$ d) $\frac{1+6c^2+3c^4}{4}$

42. If $a \sin \theta + b \cos \theta + c = 0$, $p \sin \theta + q \cos \theta + r = 0$, then $(br - qc)^2 + (cp - qr)^2 = ?$

- a) $(ap - bq)^2$ b) $(aq - bp)^2$ c) $(aq + bp)^2$ d) $(aq - bp)^3$

43. What is the value of $\frac{1}{\sin^4(90-2\alpha)} + \frac{1}{\cos^2(90-2\alpha)-1}$? SSC CHSL PRE 2024

$\frac{1}{\sin^4(90-2\alpha)} + \frac{1}{\cos^2(90-2\alpha)-1}$ का मान क्या है?

- A) $\sin^2 2\alpha + \tan^2 2\alpha$
B) $\sec^2 2\alpha \tan^2 2\alpha$
C) $\sec^2 2\alpha + \cot^2 2\alpha$
D) $\cos^2 2\alpha \cot^2 2\alpha$

44. $\sqrt{\frac{\cos \text{ec} A}{\cos \text{ec} A - 1} + \frac{\cos \text{ec} A}{\cos \text{ec} A + 1}} = 2$, then $A = ?$

- a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$
c) $\frac{\pi}{4}$ d) $\frac{\pi}{6}$

45. The value of $\left(\frac{\text{cosec} A}{\text{cosec} A - 1} + \frac{\text{cosec} A}{\text{cosec} A + 1} \right) \left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1} \right)^{-1}$, what $A = 60^\circ$ is:

$A = 60^\circ$ होने पर $\left(\frac{\text{cosec} A}{\text{cosec} A - 1} + \frac{\text{cosec} A}{\text{cosec} A + 1} \right) \left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1} \right)^{-1}$ का मान क्या होगा:

- (a) 4 (b) 3
(c) 1 (d) 2

46. If $\frac{(1 + \sin\theta - \cos\theta)}{(1 + \sin\theta + \cos\theta)} + \frac{(1 + \sin\theta + \cos\theta)}{(1 + \sin\theta - \cos\theta)} = 4$, then which of the following values will be suitable for θ ?

अगर $\frac{(1 + \sin\theta - \cos\theta)}{(1 + \sin\theta + \cos\theta)} + \frac{(1 + \sin\theta + \cos\theta)}{(1 + \sin\theta - \cos\theta)} = 4$ है, तो लिखित में से कौन सा मान θ के लिए सही होगा?

- (a) 90° (b) 60°
(c) 45° (d) 30°

47. If $117 \cos^2 A + 129 \sin^2 A = 120$ and $170 \cos^2 B + 158 \sin^2 B = 161$, then the value of $\text{cosec}^2 A \sec^2 B$ is:

यदि $117 \cos^2 A + 129 \sin^2 A = 120$ है और $170 \cos^2 B + 158 \sin^2 B = 161$ है, तो $\text{cosec}^2 A \sec^2 B$ का मान ज्ञात करें।

- (a) 1 (b) 9 (c) 4 (d) 16

48. If $\sin \theta + \sin^2 \theta = 1$ then $\cos^2 \theta + \cos^4 \theta = ?$

- a) 0 b) 1 c) -1 d) 2

49. If $\sin \theta + \sin^2 \theta = 1$ then $\cos^4 \theta + \cos^8 \theta + 2 \cos^6 \theta = ?$

- a) 0 b) 1 c) -1 d) 2

50. If $\sin \theta + \sin^2 \theta = 1$, Then $\cos^{12} \theta + 3 \cos^{10} \theta + 3 \cos^8 \theta + \cos^6 \theta - 1 = ?$

- a) 0 b) 1 c) -1 d) 2

51. If $\cos A = \sin^2 A$, and $a \sin^{12} A + b \sin^{10} A + c \sin^8 A + \sin^6 A = 1$, then $a + b + c = ?$

यदि $\cos A = \sin^2 A$ और $a \sin^{12} A + b \sin^{10} A + c \sin^8 A + \sin^6 A = 1$ है तो $a + b + c = ?$



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(a) 7

(b) 8

(c) 9

(d) 6

52. If $\cos\theta + \cos^2\theta = 1$, find the value of $\sqrt{\sin^4\theta + \cos^2\theta}$.

यदि $\cos\theta + \cos^2\theta = 1$ है, तो $\sqrt{\sin^4\theta + \cos^2\theta}$ का मान ज्ञात कीजिए।

(a) $\sqrt{2}\cos\theta$

(b) $2\cos\theta$

(c) $\sqrt{2}\sin\theta$

(d) $2\sin\theta$

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53. If $\sin A + \sin^2 A = 1$, then the value of $\cos^4 A + \cos^6 A$ is:

यदि $\sin A + \sin^2 A = 1$ है, तो $\cos^4 A + \cos^6 A$ का मान क्या होगा?

(a) 1

(b) $\cos A$

(c) $\sin A$

(d) 0

[SSC CPO 2023]

54. If $\frac{\cos\alpha}{\cos\beta} = a, \frac{\sin\alpha}{\sin\beta} = b$, then $\sin^2\beta = ?$

a) $\frac{a^2-1}{a^2+b^2}$

b) $\frac{a^2+1}{a^2-b^2}$

c) $\frac{a^2+1}{a^2+b^2}$

d) $\frac{a^2-1}{a^2-b^2}$

55. If $\frac{\cos\alpha}{\sin\beta} = 10$ and $\frac{\cos\alpha}{\cos\beta} = 11$, then the value of $\cos^2\beta$ is :

यदि $\frac{\cos\alpha}{\sin\beta} = 10$ और $\frac{\cos\alpha}{\cos\beta} = 11$ है, तो $\cos^2\beta$ का मान क्या होगा?

(a) $\frac{121}{132}$

(b) $\frac{100}{221}$

(c) $\frac{88}{108}$

(d) $\frac{221}{121}$

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56. $\tan\alpha = n\tan\beta, \sin\alpha = m\sin\beta$, then $\cos^2\alpha = ?$

a) $\frac{m^2}{n^2+1}$

b) $\frac{m^2-1}{n^2-1}$

c) $\frac{m^2+1}{n^2+1}$

d) $\frac{m^2}{n^2}$

57. If $\tan\theta = \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}$, Then $\sin\alpha + \cos\alpha = ?$

(a) $\pm\sqrt{2}\sin\theta$

(b) $\pm\sqrt{2}\cos\theta$

(c) $\pm\sqrt{2}\tan\theta$

(d) $\pm\sqrt{2}\cot\theta$

58. If $-\sin\theta + \operatorname{cosec}\theta = 6$, then what is the value of $\sin\theta + \operatorname{cosec}\theta$?

यदि $-\sin\theta + \operatorname{cosec}\theta = 6$, तो $\sin\theta + \operatorname{cosec}\theta$ का मान क्या है?

(a) 6

(b) $\sqrt{40}$

(c) $\sqrt{34}$

(d) $\sqrt{38}$

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59. If $a\sin\theta + b\cos\theta = c$ then $a\cos\theta - b\sin\theta = ?$

(a) $\pm\sqrt{a^2 + b^2 + c^2}$

(b) $\pm\sqrt{a^2 + b^2 - c^2}$

(c) $\pm\sqrt{a^2 - b^2 - c^2}$

(d) $\pm\sqrt{a^2 - b^2 + c^2}$

60. If $3\sin\theta + 5\cos\theta = 5$, then what is the value of $5\sin\theta - 3\cos\theta$ equal to?

यदि $3\sin\theta + 5\cos\theta = 5$, तो $5\sin\theta - 3\cos\theta$ का मान किसके बराबर है? [CDS 2023]

A) 5

B) -3

C) -2

D) 0



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61. If $3\sin x + 4\cos x = 2$, then the value of $3\cos x - 4\sin x$ is equal to:

यदि $3\sin x + 4\cos x = 2$ है, तो $3\cos x - 4\sin x$ का मान ज्ञात कीजिए।

- (a) $\sqrt{21}$ (b) $\sqrt{23}$ (c) 21 (d) $\sqrt{29}$

62. $\sec\theta(\cos\theta + \sin\theta) = \sqrt{2}$ then what is the value of $\frac{2\sin\theta}{\cos\theta - \sin\theta}$?

- a) $3\sqrt{2}$ b) $\frac{3}{\sqrt{2}}$ c) $\frac{1}{\sqrt{2}}$ d) $\sqrt{2}$

63. If $(a^3 + b^3)\sin\theta + (a^3 - b^3)\cos\theta = 7$ and $(a^3 + b^3)\cos\theta - (a^3 - b^3)\sin\theta = 5$, then $a^6 + b^6 = ?$

- (a) 24 (b) 37 (c) 35 (d) 49

64. If $\cos 27^\circ + \sin 27^\circ = k$, then what is the value of $\cos^2 27^\circ - \sin^2 27^\circ$?

यदि $\cos 27^\circ + \sin 27^\circ = k$ है तो $\cos^2 27^\circ - \sin^2 27^\circ$ का मान क्या है?

- (a) $k\sqrt{2 - k^2}$ (b) $k\sqrt{k^2 - 2}$ (c) $k\sqrt{1 - k^2}$ (d) $k\sqrt{k^2 - 1}$

65. If $\cos 41^\circ + \sin 41^\circ = k$, then what is the value of $\cos^3 41^\circ - \sin^3 41^\circ$?

यदि $\cos 41^\circ + \sin 41^\circ = k$ है तो $\cos^3 41^\circ - \sin^3 41^\circ$ का मान क्या है?

- (a) $\frac{(k^2 + 1)}{2} \sqrt{2 - k^2}$ (b) $\frac{(k^2 + 1)}{2} \sqrt{k^2 - 2}$
(c) $\frac{(k^2 - 1)}{2} \sqrt{2 - k^2}$ (d) $\frac{(k^2 - 1)}{2} \sqrt{k^2 - 1}$

66. If $\sin\theta - \cos\theta = \frac{4}{5}$, then find the value of $\sin\theta + \cos\theta$?

यदि $\sin\theta - \cos\theta = \frac{4}{5}$ है, तो $\sin\theta + \cos\theta$ का मान क्या होगा?

- A) $\frac{5}{\sqrt{34}}$
B) $\frac{5}{\sqrt{24}}$
C) $\frac{\sqrt{34}}{5}$
D) $\frac{\sqrt{24}}{5}$

67. If $\sin\theta + \cos\theta = \frac{\sqrt{11}}{3}$, then find $\cos\theta - \sin\theta$?

यदि $\sin\theta + \cos\theta = \frac{\sqrt{11}}{3}$, तो $\cos\theta - \sin\theta$ ज्ञात कीजिए? (CPO 2023)

- a) $\frac{\sqrt{5}}{3}$
b) $7/3$
c) $5/3$
d) $\frac{\sqrt{7}}{3}$

68. If $\sin\theta + \cos\theta = \frac{23}{17}$, Then $\sin\theta - \cos\theta = ?$

- (a) $\pm \frac{15}{17}$ (b) $\pm \frac{9}{17}$ (c) $\pm \frac{6}{17}$ (d) $\pm \frac{7}{17}$

69. If $\sin\theta - \cos\theta = \frac{1}{29}$ find the value of $\sin\theta + \cos\theta$.

यदि $\sin\theta - \cos\theta = 1/29$ है तो $\sin\theta + \cos\theta$ का मान ज्ञात करें।

- (a) $\frac{41}{29}$ (b) $\frac{42}{29}$ (c) $\frac{22}{29}$ (d) $\frac{2}{29}$

70. $12\sin\theta + 35\cos\theta = 37$, then $35\sin\theta - 12\cos\theta = ?$

- (a) 0 (b) 1 (c) 37 (d) 12

71. $24\sin\theta + 7\cos\theta = 25$, then $7\sin\theta + 24\cos\theta = ?$

- (a) 25 (b) $336/25$ (c) $317/24$ (d) $275/7$

72. If $11\sin\theta + 60\cos\theta = 61$; $0 < \theta < 90^\circ$, then what is the value of $\sqrt{660(\tan\theta + \cot\theta)}$?

यदि $11\sin\theta + 60\cos\theta = 61$; $0 < \theta < 90^\circ$, तो $\sqrt{660(\tan\theta + \cot\theta)}$ का मान क्या है?

- A) 61
B) $61\sqrt{2}$
C) 122
D) $122\sqrt{2}$

73. $(a^2 - b^2)\sin\theta + 2ab\cos\theta = a^2 + b^2$, then $\tan\theta = ?$

- (a) $\frac{a^2 - b^2}{2ab}$ (b) $\frac{2ab}{a^2 - b^2}$ (c) $\frac{a^2 - b^2}{a^2 + b^2}$ (d) $\frac{a^2 + b^2}{a^2 - b^2}$

74. If $\frac{\sin^4 A}{a} + \frac{\cos^4 A}{b} = \frac{1}{a+b}$, Then $\frac{\sin^8 A}{a^3} + \frac{\cos^8 A}{b^3} = ?$



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(a) $\frac{1}{(a+b)^2}$
(c) $\frac{1}{(a+b)^3}$

(b) $\frac{a^2 b^2}{(a+b)^2}$
(d) $\frac{a^3 b^3}{(a+b)^2}$

75. If $x \cos \theta - y \sin \theta = \sqrt{x^2 + y^2}$ and $\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} = \frac{1}{x^2 + y^2}$, then the correct relation is:

a) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ b) $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ d) $\frac{x^2}{b^2} - \frac{y^2}{a^2} = 1$

76. If $24 \sin^4 \alpha + 40 \cos^4 \alpha = 15$, Then $54 \tan^6 \alpha + 125 \operatorname{cosec}^6 \alpha = ?$

(a) 729 (b) 762 (c) 716 (d) 625

77. If $\frac{\cos^4 \alpha}{\cos^2 \beta} + \frac{\sin^4 \alpha}{\sin^2 \beta} = 1$, then $\frac{\cos^4 \beta}{\cos^2 \alpha} + \frac{\sin^4 \beta}{\sin^2 \alpha} = ?$

(a) 1 (b) 2 (c) 0 (d) -1

78. The value of $\frac{\sec \theta (1 - \sin \theta)(\sin \theta + \cos \theta)(\sec \theta + \tan \theta)}{\sin \theta (1 + \tan \theta) + \cos \theta (1 + \cot \theta)}$ is equal to

$\frac{\sec \theta (1 - \sin \theta)(\sin \theta + \cos \theta)(\sec \theta + \tan \theta)}{\sin \theta (1 + \tan \theta) + \cos \theta (1 + \cot \theta)}$ का मान बराबर है :

a) $\operatorname{cosec} \theta \sec \theta$ b) $\sin \theta \cos \theta$
c) $2 \cos \theta$ d) $2 \sin \theta$

79. $\sin^4 \theta + \cos^4 \theta$ in terms of $\sin \theta$ can be written as:

$\sin^4 \theta + \cos^4 \theta$ को $\sin \theta$ के रूप में लिखा जा सकता है:

A) $2 \sin^4 \theta + 2 \sin^2 \theta - 1$
B) $2 \sin^4 \theta - 2 \sin^2 \theta$
C) $2 \sin^4 \theta - 2 \sin^2 \theta - 1$
D) $2 \sin^4 \theta - 2 \sin^2 \theta + 1$

80. If $l \cos^2 \theta + m \sin^2 \theta = \frac{\cos^2 \theta (\operatorname{cosec}^2 \theta + 1)}{\operatorname{cosec}^2 \theta - 1}$, (where $0^\circ < \theta < 90^\circ$), then find the value of $\tan \theta$?

यदि $l \cos^2 \theta + m \sin^2 \theta = \frac{\cos^2 \theta (\operatorname{cosec}^2 \theta + 1)}{\operatorname{cosec}^2 \theta - 1}$, (where $0^\circ < \theta < 90^\circ$) है, तो $\tan \theta$ का मान ज्ञात कीजिये?

A) $\sqrt{\frac{l-2}{l-m}}$ C) $\sqrt{\frac{2-l}{l-m}}$
B) $\sqrt{\frac{l-2}{m-1}}$ D) $\sqrt{\frac{l-1}{2-m}}$

81. The value of expression $\frac{1 - \sin(2t)}{1 + \sin(2t)} \times \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)}$ is:

व्यंजक $\frac{1 - \sin(2t)}{1 + \sin(2t)} \times \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)}$ का मान है:

(a) $\frac{1 - 2 \tan(t)}{1 + 2 \tan(t)}$ (b) $\frac{1 - \tan(t)}{1 + \tan(t)}$
(c) $\frac{1 + 2 \tan(t)}{1 - 2 \tan(t)}$ (d) $\frac{1 + \tan(t)}{1 - \tan(t)}$

82. If $\theta = 53^\circ$ then $\frac{1}{2} \sqrt{1 + \sin \theta} + \sqrt{1 - \sin \theta}$ is equal to?

यदि $\theta = 53^\circ$, तो $\frac{1}{2} \sqrt{1 + \sin \theta} + \sqrt{1 - \sin \theta}$ किसके बराबर है?

(a) $\cot \frac{\theta}{2}$ (b) $\sec \frac{\theta}{2}$ (c) $\sin \frac{\theta}{2}$ (d) $\cos \frac{\theta}{2}$